

A bisection-width Crossing Lemma for regular spectral expanders, with an Albertson corollary

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Abstract

We package the Pach–Spencer–Tóth bisection-width Crossing Lemma together with Alon’s spectral bisection bound for regular spectral expanders into a single explicit inequality. For every d_0 -regular graph G on n vertices with second adjacency eigenvalue $|\lambda_2(G)| \leq \theta d_0$,

$$\text{cr}(G) \geq \frac{(1 - \theta)^2 d_0^2 n^2}{1280} - \frac{d_0^2 n}{16}.$$

(The constant $1/1280$ comes from a self-contained inline derivation through the dual bisection inequality of Pach–Shahrokhi–Szegedy / Sýkora–Vrt’o. Pach–Spencer–Tóth sharpen the underlying bisection-cum-crossing constant from $1/80$ to $1/40$; using their sharper form replaces $1/1280$ by $1/640$ throughout. See Remark 8 for the trade-off.) Both inputs are classical (Pach–Spencer–Tóth 2000, Alon 1986 via the expander mixing lemma); the contribution here is the explicit packaging of the spectral parameter θ inside the Crossing-Lemma constant, which to our knowledge has not been written down in this form. We then deduce an Albertson-type corollary: for every t -list-critical d_0 -regular spectral- θ -expander G with $d_0 \geq t - 1$ and $\theta \leq 1 - 5\sqrt{2}/(d_0\sqrt{n}) \cdot \sqrt{Z(t)}/\sqrt{n}$ (an explicit smallness condition; see Corollary 9 for the precise form), $\text{cr}(G) \geq \text{cr}(K_t)$.

We are explicit about the scope of this paper. The result does *not* address the Cranston residual triples $(t, n) \in \{(25, 48), (26, 50), (26, 51)\}$: only $(26, 51)$ admits Ore candidates (by the congruence $|V| \equiv 1 \pmod{k-1}$ for k -Ore graphs), and at that one Ore corner the candidate graphs are emphatically *not* spectral expanders, with $\theta = 1 - O(1/n)$ even as $d_0 \rightarrow \infty$. For the other two residual triples, the present theorem says nothing about the minimum-counterexample candidates either. We are not aware of an earlier Crossing-Lemma inequality with explicit spectral dependence; the present packaging makes the dependence available for use.

1 Introduction

The Crossing Lemma family. The Crossing Lemma of Ajtai, Chvátal, Newborn, Szemerédi [2] and Leighton [18] states that for every graph G with $n = |V(G)|$ and $m = |E(G)| \geq 4n$,

$$\text{cr}(G) \geq \frac{c m^3}{n^2}, \tag{1}$$

for an absolute constant $c > 0$. Their original argument gives $c = 1/64$. Successive improvements have driven the constant down: Pach and Tóth [25] obtained $c = 1/33.75$ via two k -planarity bounds; Pach, Radoičić, Tardos, and Tóth [26] improved this to $c = 1/31.1$; Ackerman [1] obtained $c = 1/29$ via a $|E| \leq 6n - 12$ bound for graphs with at most four crossings per edge; and Bungener and Kaufmann [6] have recently improved the constant to $c = 1/27.48$ under a slightly larger density threshold $m > 6.77n$.

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The bisection-width Crossing Lemma family. Parallel to the density-iteration line, Leighton [19] introduced a bisection-width variant of the Crossing Lemma that gives lower bounds on $\text{cr}(G)$ controlled by the bisection width $\text{bw}(G)$ (the minimum number of edges crossing any balanced bipartition of $V(G)$). Pach, Shahrokhi, and Szegedy [23] and Sýkora and Vrt'o [28] established the converse-direction inequality

$$\text{bw}(G) \leq 6.32 \sqrt{\text{cr}(G)} + 1.58 \sqrt{\sum_v \deg(v)^2}, \quad (2)$$

from which we derive in Section 3 the self-contained inequality

$$\text{cr}(G) \geq \frac{\text{bw}(G)^2}{80} - \frac{\sum_v \deg(v)^2}{16}. \quad (3)$$

Pach, Spencer, and Tóth [24] sharpen the constant $1/80$ to $1/40$ by a more careful split of the dual bound (2); their sharper form is what they use to conclude $\text{cr}(G(n, p)) \geq e^2/4000$ on Erdős–Rényi graphs. We work with (3) throughout for self-containment (see Remark 8).

The spectral input. For d_0 -regular graphs with second adjacency eigenvalue $|\lambda_2| \leq \theta d_0$, the expander mixing lemma [4, 5] gives

$$\text{bw}(G) \geq \frac{(1 - \theta) d_0 n}{4}. \quad (4)$$

Substituting (4) into (3) and using $\sum_v \deg(v)^2 = d_0^2 n$ for d_0 -regular graphs gives the main theorem of this note.

Theorem 1. *Let G be a d_0 -regular graph on n vertices with second adjacency eigenvalue $|\lambda_2(G)| \leq \theta d_0$ for some $\theta \in [0, 1)$. Then*

$$\text{cr}(G) \geq \frac{(1 - \theta)^2 d_0^2 (\lfloor n/2 \rfloor \lceil n/2 \rceil)^2}{80 n^2} - \frac{d_0^2 n}{16}. \quad (5)$$

For n even, $\lfloor n/2 \rfloor \lceil n/2 \rceil = n^2/4$, so

$$\text{cr}(G) \geq \frac{(1 - \theta)^2 d_0^2 n^2}{1280} - \frac{d_0^2 n}{16}.$$

For n odd, the right-hand side of (5) equals $(1 - 1/n^2)^2 \cdot (1 - \theta)^2 d_0^2 n^2 / 1280 - d_0^2 n / 16$, i.e. the clean even- n form weakened by a factor $(1 - 1/n^2)^2 \geq (1 - 2/n^2)$. The constant $1/80$ in (5) can be replaced by $1/40$ – halving the leading denominator – if one invokes the sharper Pach–Spencer–Tóth form of (3) (Remark 8).

What is new and what is not. Both ingredients of Theorem 1 are classical. The Pach–Spencer–Tóth inequality (3) was published in 2000 [24]; the spectral bisection bound (4) is a corollary of the expander mixing lemma of Alon and Chung [5], itself a consequence of Alon’s eigenvalue analysis [4]. The contribution of the present note is simply to write down the explicit packaging of the spectral parameter θ inside the Crossing-Lemma constant. To our knowledge, no Crossing-Lemma inequality in the literature gives an explicit dependence on θ ; the PST inequality (3) is stated in terms of $\text{bw}(G)$, and the conversion to spectral data has not been recorded.

The motivation comes from a separate line of work on Albertson’s conjecture [3]: on critical graphs whose minimum degree saturates the Dirac bound $\delta(G) \geq t - 1$, the standard Crossing-Lemma constant $1/27.48$ is too weak to certify $\text{cr}(G) \geq \text{cr}(K_t)$ in the Cranston residual range $t \in \{25, 26\}$ [7]; a Crossing Lemma with explicit spectral dependence opens up the possibility of using the expander structure of certain critical-graph candidates. In Section 5 we record the corresponding Albertson corollary; in Section 6 we are explicit that this does *not* close the Cranston residual, because the residual candidate graphs (Ore compositions of K_{26}) are not spectral expanders.

Organisation. Section 2 fixes notation and reviews bisection width, the spectral gap, and the expander mixing lemma. Section 3 re-derives the PST bisection-width Crossing Lemma (3) with explicit constants from the dual bound (2). Section 4 assembles Theorem 1 and verifies its numerical content at one Ramanujan regime. Section 5 states and proves the Albertson corollary. Section 6 records the limitations, in particular the failure of the result on the Cranston residual.

2 Preliminaries

We work with finite simple graphs throughout. Standard graph-theory notation follows [11]. The *crossing number* $\text{cr}(G)$ is the minimum number of pairwise edge-crossings over all drawings of G in the plane.

Definition 2 (Bisection width). The *bisection width* of a graph G on n vertices is

$$\text{bw}(G) := \min\{|E(A, B)| : V(G) = A \sqcup B, \lfloor n/2 \rfloor \leq |A|, |B| \leq \lceil n/2 \rceil\},$$

where $E(A, B)$ denotes the set of edges of G with one endpoint in A and one in B .

We follow the convention of Pach–Spencer–Tóth [24] and require that the bipartition be *balanced*. Some authors (e.g. [20]) instead require only that $|A|, |B| \geq n/3$; the two conventions differ by at most a factor of 2 in the resulting $\text{bw}(G)$ on regular graphs, which is absorbed into the implicit constants below.

Definition 3 (Spectral expansion). Let G be a d -regular graph on n vertices and let $A_G \in \mathbb{R}^{n \times n}$ be its adjacency matrix. The eigenvalues of A_G are real, $d = \lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_n \geq -d$. We write $\lambda_2(G) := \max(|\lambda_2|, |\lambda_n|)$ for the *second adjacency eigenvalue*, and call G a θ -*spectral expander* (for $\theta \in [0, 1)$) if $\lambda_2(G) \leq \theta d$.

The fundamental tool relating spectrum to combinatorics is the expander mixing lemma:

Lemma 4 (Expander mixing lemma; Alon–Chung [5]). *Let G be a d -regular graph on n vertices with second adjacency eigenvalue $\lambda_2(G) \leq \theta d$. For every pair of disjoint sets $S, T \subseteq V(G)$,*

$$\left| e(S, T) - \frac{d|S||T|}{n} \right| \leq \theta d \sqrt{|S||T|(1 - |S|/n)(1 - |T|/n)}. \quad (6)$$

Corollary 5 (Spectral bisection bound). *Let G be a d -regular graph on n vertices with $\lambda_2(G) \leq \theta d$. Then*

$$\text{bw}(G) \geq (1 - \theta) \frac{d \lfloor n/2 \rfloor \lceil n/2 \rceil}{n}.$$

In particular, $\text{bw}(G) \geq (1 - \theta)dn/4$ when n is even, and $\text{bw}(G) \geq (1 - \theta)(1 - 1/n^2)dn/4$ when n is odd.

Proof. Let $V(G) = A \sqcup B$ be a balanced bipartition, so $\{|A|, |B|\} = \{\lfloor n/2 \rfloor, \lceil n/2 \rceil\}$. Set $s = \lfloor n/2 \rfloor$ and $t = \lceil n/2 \rceil$, so $s + t = n$. By Lemma 4 with $S = A$, $T = B$,

$$e(A, B) \geq \frac{dst}{n} - \theta d \sqrt{st(1 - s/n)(1 - t/n)} = \frac{dst}{n} - \theta d \frac{st}{n} = (1 - \theta) \frac{dst}{n},$$

where we used $(1 - s/n)(1 - t/n) = (t/n)(s/n) = st/n^2$. For n even, $s = t = n/2$, so $st/n = n/4$ and we recover the headline bound. For n odd, $s = (n - 1)/2$ and $t = (n + 1)/2$, so $st = (n^2 - 1)/4$ and $st/n = (n^2 - 1)/(4n) = (n/4)(1 - 1/n^2)$. $\square$

Corollary 5 is folklore. The constant $1/4$ is sharp at $\theta = 0$ (e.g. on $K_{n,n}$, $d = n/2$, $\lambda_2 = 0$, and $\text{bw}(K_{n,n}) = n^2/4 = d n/2$, exceeding the bound). For explicit references, see [15, Theorem 2.4] for a textbook account of the expander mixing lemma and its bisection-width consequence.

Lemma 6 (Ramanujan bound; Alon–Boppana [4, 21]). *Every d -regular graph on n vertices satisfies $\lambda_2(G) \geq 2\sqrt{d-1} - o_n(1)$. Equivalently, the best possible spectral expansion of an infinite family of d -regular graphs is $\theta = 2\sqrt{d-1}/d$. Graphs achieving this bound asymptotically are called Ramanujan graphs; explicit constructions exist for $d = p + 1$ with p prime [17].*

3 A self-contained bisection-width Crossing Lemma

We derive (3) inline from the dual bisection inequality of Pach–Shahrokhi–Szegedy and Sýkora–Vrt’o.

Lemma 7 (Bisection-width Crossing Lemma; conservative form). *Every graph G on n vertices satisfies*

$$\text{cr}(G) \geq \frac{\text{bw}(G)^2}{80} - \frac{\sum_{v \in V(G)} \deg(v)^2}{16}. \quad (7)$$

Proof. Pach–Shahrokhi–Szegedy [23] and Sýkora–Vrt’o [28] independently established the dual estimate (2):

$$\text{bw}(G) \leq 6.32 \sqrt{\text{cr}(G)} + 1.58 \sqrt{\sum_v \deg(v)^2}.$$

The two coefficients satisfy $1.58 = 6.32/4$ exactly, so the right-hand side factors as

$$6.32 \left(\sqrt{\text{cr}(G)} + \frac{1}{4} \sqrt{\sum_v \deg(v)^2} \right).$$

Squaring and using $(a + b)^2 \leq 2a^2 + 2b^2$ on the inner sum gives

$$\text{bw}(G)^2 \leq 2 \cdot (6.32)^2 \left(\text{cr}(G) + \frac{1}{16} \sum_v \deg(v)^2 \right) = 79.8848 \cdot \left(\text{cr}(G) + \frac{1}{16} \sum_v \deg(v)^2 \right).$$

Dividing by 79.8848 and rearranging,

$$\text{cr}(G) \geq \frac{\text{bw}(G)^2}{79.8848} - \frac{1}{16} \sum_v \deg(v)^2 \geq \frac{\text{bw}(G)^2}{80} - \frac{\sum_v \deg(v)^2}{16},$$

where the second inequality uses $79.8848 \leq 80$ to round the leading denominator upward (the second term is unchanged because the factor $1/16$ is exact, not approximate). $\square$

Remark 8. The conservative proof above gives the constant $1/80$ on the $\text{bw}(G)^2$ term. Pach, Spencer, and Tóth [24, Section 4] sharpen this to $1/40$ by a more careful split of the dual inequality (2) – allocating the two right-hand-side contributions optimally before squaring, rather than using the $(a + b)^2 \leq 2a^2 + 2b^2$ bookkeeping step in the proof above. Their proof yields

$$\text{cr}(G) \geq \frac{\text{bw}(G)^2}{40} - \frac{\sum_v \deg(v)^2}{16}.$$

Substituting this sharper $1/40$ in place of $1/80$ everywhere in the proof of Theorem 1 replaces the constant $1/1280$ by $1/640$ in (5), doubling the leading expander coefficient. We work with the self-contained $1/80$ form throughout the remainder of this note; the headline statement of Theorem 1 therefore uses $1/1280$, and the corollary constants in Section 5 are derived accordingly.

4 Proof and numerical illustration

Proof of Theorem 1. Let G be d_0 -regular on n vertices with $\lambda_2(G) \leq \theta d_0$. By Lemma 7,

$$\text{cr}(G) \geq \frac{\text{bw}(G)^2}{80} - \frac{\sum_v \deg(v)^2}{16} = \frac{\text{bw}(G)^2}{80} - \frac{d_0^2 n}{16},$$

where the last equality uses $\sum_v \deg(v)^2 = d_0^2 n$ for d_0 -regular G . By Corollary 5, $\text{bw}(G) \geq (1 - \theta) d_0 \lfloor n/2 \rfloor \lceil n/2 \rceil / n$, so

$$\text{bw}(G)^2 \geq \frac{(1 - \theta)^2 d_0^2 (\lfloor n/2 \rfloor \lceil n/2 \rceil)^2}{n^2}.$$

Substituting into the displayed inequality yields the headline form (5). For n even, the floor product equals $n^2/4$ and the numerator collapses to $(1 - \theta)^2 d_0^2 n^2 / 16$, giving the $(1 - \theta)^2 d_0^2 n^2 / 1280$ leading term. For n odd, the floor product equals $(n^2 - 1)/4$, so the leading term becomes

$$\frac{(1 - \theta)^2 d_0^2 (n^2 - 1)^2}{80 \cdot 16 \cdot n^2} = (1 - 1/n^2)^2 \cdot \frac{(1 - \theta)^2 d_0^2 n^2}{1280},$$

i.e. the even- n form weakened by the factor $(1 - 1/n^2)^2$. $\square$

Numerical sanity check at a Ramanujan regime. We illustrate the strength of Theorem 1 at one concrete spectral-expander regime. Take $d_0 = 14$ and let G be a 14-regular Ramanujan graph (so $\lambda_2(G) = 2\sqrt{13} \approx 7.21$, giving $\theta = 2\sqrt{13}/14 \approx 0.515$, and $(1 - \theta)^2 \approx 0.235$). With $d_0 = 14$ we have $m/n = 7 > 6.95$, so the Bungenier–Kaufmann threshold [6] is met. Compute at three values of n :

- $n = 1000$ (even): $\text{bw} \geq (1 - \theta) \cdot 14 \cdot 250 = 3500(1 - \theta)$, so Theorem 1 gives $\text{cr}(G) \geq 0.235 \cdot 14^2 \cdot 10^6 / 1280 - 14^2 \cdot 1000 / 16 \approx 36,000 - 12,250 \approx 23,750$.
- The Bungenier–Kaufmann Crossing Lemma at the same $(n, d_0) = (1000, 14)$ gives $m = 7000$ and $\text{cr}(G) \geq m^3 / (27.48 n^2) = 3.43 \cdot 10^{11} / (2.748 \cdot 10^7) \approx 12,500$.
- At $n = 10,000$: Theorem 1 gives $0.235 \cdot 196 \cdot 10^8 / 1280 - 196 \cdot 10^4 / 16 \approx 3.6 \cdot 10^6 - 1.2 \cdot 10^5 \approx 3.5 \cdot 10^6$. BK gives $7^3 \cdot 10^{12} / (27.48 \cdot 10^8) \approx 1.25 \cdot 10^6$.

At $(n, d_0, \theta) = (1000, 14, 0.515)$, the headline conservative form of Theorem 1 is roughly $1.9\times$ stronger than BK; at $n = 10,000$ it is roughly $2.8\times$ stronger. Using the sharper PST $1/40$ constant (Remark 8) doubles the leading term and roughly doubles the gap. The improvement grows as n/d_0 grows (because the bisection-derived bound scales as $d_0^2 n^2$, while the BK bound on a d_0 -regular graph gives $\text{cr} \gtrsim d_0^3 n$, smaller by a factor n/d_0).

A remark on the density threshold. The comparison with [6] above requires $m \geq 6.95 n$, i.e. $d_0 \geq 14$ for d_0 -regular graphs. For smaller $d_0 \in \{4, 5, \dots, 13\}$, the appropriate density-based Crossing Lemma is either the classical ACNS [2, 18] $\text{cr} \geq m^3 / (64 n^2)$ (applicable for $m \geq 4n$, i.e. $d_0 \geq 8$) or the Pach–Tóth [25] $1/33.75$ form. The qualitative n/d_0 -improvement-factor analysis carries over.

Density threshold. The bound (5) of Theorem 1 is non-trivial (positive right-hand side) exactly when

$$\frac{(1 - \theta)^2 d_0^2 n^2}{1280} > \frac{d_0^2 n}{16}, \quad \text{i.e.,} \quad n > \frac{80}{(1 - \theta)^2}.$$

For $\theta = 0.6$ this gives $n > 500$; for $\theta = 0$ (complete-bipartite-like spectrum), $n > 80$. Below this threshold the second term dominates and the inequality is vacuous. Using the sharper PST $1/40$ constant (Remark 8) halves these thresholds.

5 The Albertson corollary

We now record an Albertson-type corollary of Theorem 1 on the class of t -critical d_0 -regular spectral expanders.

Setup. Recall Guy's formula [14] for the conjectured crossing number of the complete graph K_t ,

$$\text{cr}(K_t) \leq Z(t) := \frac{1}{4} \lfloor t/2 \rfloor \lfloor (t-1)/2 \rfloor \lfloor (t-2)/2 \rfloor \lfloor (t-3)/2 \rfloor.$$

The conjecture is open for $t \geq 13$, but the upper bound $\text{cr}(K_t) \leq Z(t)$ is unconditional (realised by Zarankiewicz's drawing). We have $Z(t) \leq t^4/64$.

Recall also Dirac's theorem [12]: every t -critical graph G satisfies $\delta(G) \geq t-1$. The corresponding list-critical bound ([10, Lemma 2.1 of our companion list paper]) gives the same conclusion for t -list-critical G .

Corollary 9 (Albertson for regular spectral-expander critical graphs). *Let $t \geq 4$ and let G be a t -critical (or t -list-critical) d_0 -regular graph on n vertices with $\lambda_2(G) \leq \theta d_0$. Suppose*

(a) $d_0 \geq t-1$ (automatic from the Dirac bound on any t -critical graph),

(b) $(1-\theta)^2 \geq 1280(Z(t) + d_0^2 n/16)/(d_0^2 n^2)$, equivalently

$$(1-\theta)^2 d_0^2 n^2 \geq 1280 Z(t) + 80 d_0^2 n. \quad (8)$$

Then $\text{cr}(G) \geq \text{cr}(K_t)$.

Proof. Theorem 1 gives $\text{cr}(G) \geq (1-\theta)^2 d_0^2 n^2 / 1280 - d_0^2 n / 16$. Hypothesis (b) is exactly the inequality

$$\frac{(1-\theta)^2 d_0^2 n^2}{1280} \geq Z(t) + \frac{d_0^2 n}{16},$$

which rearranges to $\text{cr}(G) \geq Z(t) \geq \text{cr}(K_t)$. $\square$

Simplified asymptotic form. For large t and large n , condition (8) simplifies. Using $Z(t) \leq t^4/64$ and $d_0 \geq t-1 \geq t/2$ (the latter for $t \geq 2$),

$$\frac{1280 Z(t)}{d_0^2 n^2} \leq \frac{1280 t^4/64}{(t/2)^2 n^2} = \frac{80 t^2}{n^2},$$

so (8) is implied by $(1-\theta)^2 \geq 80 t^2/n^2 + 80/n$, equivalently

$$\theta \leq 1 - \sqrt{80 t^2/n^2 + 80/n}. \quad (9)$$

For $n \gtrsim 10t$, the term $80 t^2/n^2 \leq 80/100 = 0.80$ already gives $\theta \leq 1 - 0.90 \approx 0.10$; for $n \gtrsim 20t$, the bound becomes $\theta \leq 1 - 0.32$, so any $\theta \leq 0.68$ suffices. This is well within Ramanujan range $\theta = 2\sqrt{d_0-1}/d_0$ for $d_0 \geq 10$ (where $\theta_{\text{Ramanujan}} \approx 0.6$).

Smallest t where the corollary is non-vacuous. For $t = 4$, we have $\text{cr}(K_4) = 0$, so the inequality is trivial. For $t = 5$, $\text{cr}(K_5) = 1$, also easily met. The first non-trivial case is $t = 7$ ($\text{cr}(K_7) = 9$), or $t = 13$ if one demands $\text{cr}(K_t)$ matching $Z(t)$ unconditionally (the regime where Albertson is genuinely open). For $t = 13$, $Z(13) = 225$, so (8) requires $(1-\theta)^2 d_0^2 n^2 \geq 288000 + 80 d_0^2 n$. At $d_0 = 12$ and $n = 60$, the left-hand side is $(1-\theta)^2 \cdot 144 \cdot 3600 = 518400(1-\theta)^2$ and the right-hand side is $288000 + 80 \cdot 144 \cdot 60 = 288000 + 691200 = 979200$, exceeding the left-hand side even at $\theta = 0$. At $d_0 = 12$ and $n = 120$ the left-hand side is $(1-\theta)^2 \cdot 144 \cdot 14400 =$

$2073600(1-\theta)^2$ and the right-hand side is $288000 + 80 \cdot 144 \cdot 120 = 288000 + 1382400 = 1670400$, giving $(1-\theta)^2 \geq 1670400/2073600 \approx 0.806$, i.e. $\theta \leq 1 - 0.898 \approx 0.102$. The Ramanujan bound for $d_0 = 12$ is $\theta = 2\sqrt{11}/12 \approx 0.553$, so even optimal expanders at this regime do not meet the corollary's hypothesis under the conservative $1/1280$ constant; using the sharper PST $1/40$ form (Remark 8) halves the coefficients in (8) and brings the threshold down to $\theta \leq 0.398$, which a Ramanujan graph at $d_0 = 12, n = 120$ already meets. The corollary becomes non-vacuous for Ramanujan expanders once $n \gtrsim 30t$ under the conservative form ($n \gtrsim 15t$ under the sharper PST form), which is well above the Cranston regime $n \approx 2t$.

This is the principal honest qualification: the corollary produces a Crossing-Lemma improvement on *moderately dense* expander critical graphs ($n \gtrsim 20t$), not on the tight Dirac-edge-floor regime ($n \approx 2t$) that the Albers conjecture is concerned with. We elaborate on this point in Section 6.

6 Limitations and scope

6.1 The Cranston residual is untouched

The triples $(t, n) \in \{(25, 48), (26, 50), (26, 51)\}$ at which the unconditional Albers conjecture is currently open [7] include exactly one ‘‘Ore corner’’: only $(t, n) = (26, 51)$ admits Ore compositions of K_{26} [22], because the order of any k -Ore graph satisfies the congruence $|V| \equiv 1 \pmod{k-1}$. The remaining two triples $(25, 48)$ and $(26, 50)$ have no Ore candidates, since $48 \not\equiv 1 \pmod{24}$ and $50 \not\equiv 1 \pmod{25}$. At the Ore corner $(26, 51)$, the construction takes two copies of K_{26} joined by a small ‘‘interface’’ via edge-identification on a small common subgraph; the resulting graph is 25-regular on $n = 51$ vertices, with edge count meeting the Kostochka–Yancey [16] bound $m \geq \lceil ((t+1)(t-2)n - t(t-3))/(2(t-1)) \rceil = 649$.

Ore graphs are not spectral expanders. The adjacency matrix of an Ore composition $G = K_{26} * K_{26}$ at the Ore corner has a clear block-bipartition structure: viewing $V(G)$ as $A \sqcup B$ with $|A| = 26, |B| = 25$ (or vice versa), the within-block edges (each block is nearly K_{26}) contribute eigenvalues bounded by the K_{26} spectrum, but the small inter-block interface produces a near-zero second eigenvalue *only relative to the $K_{n,n}$ -like structure*. Concretely, a near-balanced bipartition of G via (A, B) has very few cut edges (the $\Theta(1)$ interface), so

$$\text{bw}(G) = O(1) \quad (\text{not } \Omega(d_0 n) \text{ as for expanders}).$$

By Corollary 5 in reverse, this forces θ to be *very close to 1*: if $\text{bw}(G) \leq c$ with $c = O(1)$, then $(1-\theta)d_0 \lfloor n/2 \rfloor \lceil n/2 \rceil / n \leq c$, giving $1-\theta \leq 4c/(d_0(n-1/n)) = O(1/n)$. Hence $\theta = 1 - O(1/n)$, and the constant $(1-\theta)^2$ in (5) is $O(1/n^2)$, making the bound vacuous.

In particular, applying Theorem 1 to the Ore corner graph $(n, d_0) = (51, 25)$ with $\theta = 1 - O(1/51)$ gives $\text{cr}(G) \geq O(1) - 25^2 \cdot 51/16 = O(1) - 1992$, which is not useful. For the non-Ore triples $(25, 48)$ and $(26, 50)$, the relevant minimum counterexample candidates are not Ore compositions, and the present theorem says nothing about them either.

6.2 Open: non-expander reduction

The natural further question is whether any t -critical non-expander can be ε -perturbed into an expander preserving criticality. Possibilities:

- *Cayley-graph cover.* Replace G by a Cayley graph Γ on a group Γ acting on $V(G)$, with the generating set chosen so that Γ is a spectral expander and admits G as a quotient. If Γ inherits t -criticality from G , Theorem 1 applies to Γ , giving $\text{cr}(\Gamma) \geq \text{explicit}$. By the standard minor-monotonicity of crossing number, $\text{cr}(\Gamma) \geq \text{cr}(G)$ on subgraphs, but

the ε -perturbation goes in the wrong direction here (adding edges to G to produce Γ does not give $\text{cr}(G) \geq \text{cr}(\Gamma)$).

- *Lifted-product construction.* The recent lifted-product constructions of quantum LDPC codes [27] produce explicit expanders from non-expander inputs by a topological lift. The applicability to preserve t -criticality is not, to our knowledge, addressed in the literature.

We do not pursue these directions here; they are recorded as the 12-month speculative extension noted in our attack memo [9].

6.3 Relation to the companion R5a and R3.6 papers

This paper is the third in a 12-month bundle of partial-result papers on Albertson’s conjecture. The companion notes are:

- [8]: *An algebraic explanation for the degree threshold $9/8$ in Fox–Pach–Suk’s bound towards Albertson’s conjecture.* Establishes sharpness of the constant $9/16$ in [13, Lemma 2.3] within the FPS Vizing–Gupta plus semi-random framework.
- [10]: *List-coloring Albertson up to chromatic number 18.* Lifts the Albertson–Cranston–Fox / Barát–Tóth / Ackerman chain to list-coloring at the Ackerman threshold $t \leq 18$.
- This paper: *A bisection-width Crossing Lemma for regular spectral expanders.* Records the explicit- θ packaging of PST plus Alon and the corresponding Albertson corollary.

None of the three closes the Cranston residual at $t \in \{25, 26\}$. Each is a stand-alone graph-theory result that addresses one of the three identifiable structural slacks in the Albertson chain: the FPS chromatic-index constant ([8]), the chromatic-versus-list-chromatic boundary ([10]), and the density-vs-expander boundary in the Crossing-Lemma constant (this paper).

Acknowledgements

This paper grew out of an attack memo [9] attempting a direct min-degree-aware refinement of the Bungenier–Kaufmann Crossing Lemma. That attack failed cleanly — the density-iteration machinery of all known Crossing-Lemma proofs is degree-sequence insensitive — and the present paper is the proven fallback. We thank the broader Albertson team for collaboration on the bundle of three companion papers.

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